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Autonomous AI Retail Decision Intelligence

Theme: AI FOR BUSINESS

TEAM: MINDSPARKS

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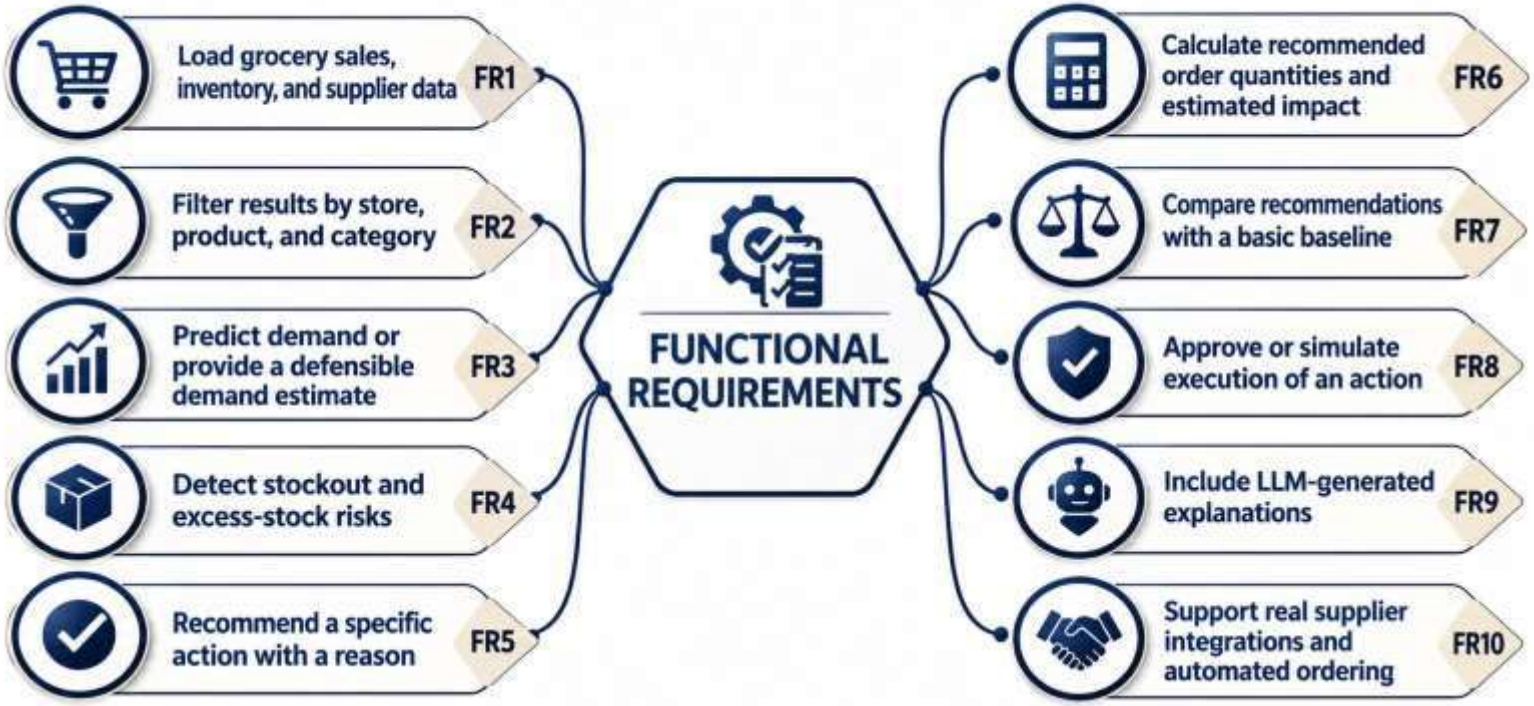
ABSTRACT

Problem Statement

- **Demand Prediction:** Difficulty predicting customer demand across different stores and products.
- **Inventory Management:** Facing Stock shortages, excess stock and product expiry.
- **Replenishment & Logistics:** Challenges in deciding when, how much and where to deliver products.
- **Profit Management:** Discounts, rising costs and poor decisions reducing profits.
- **Decision-Making:** Lack of connection to improve business operations and customer satisfaction.

Target Avenues





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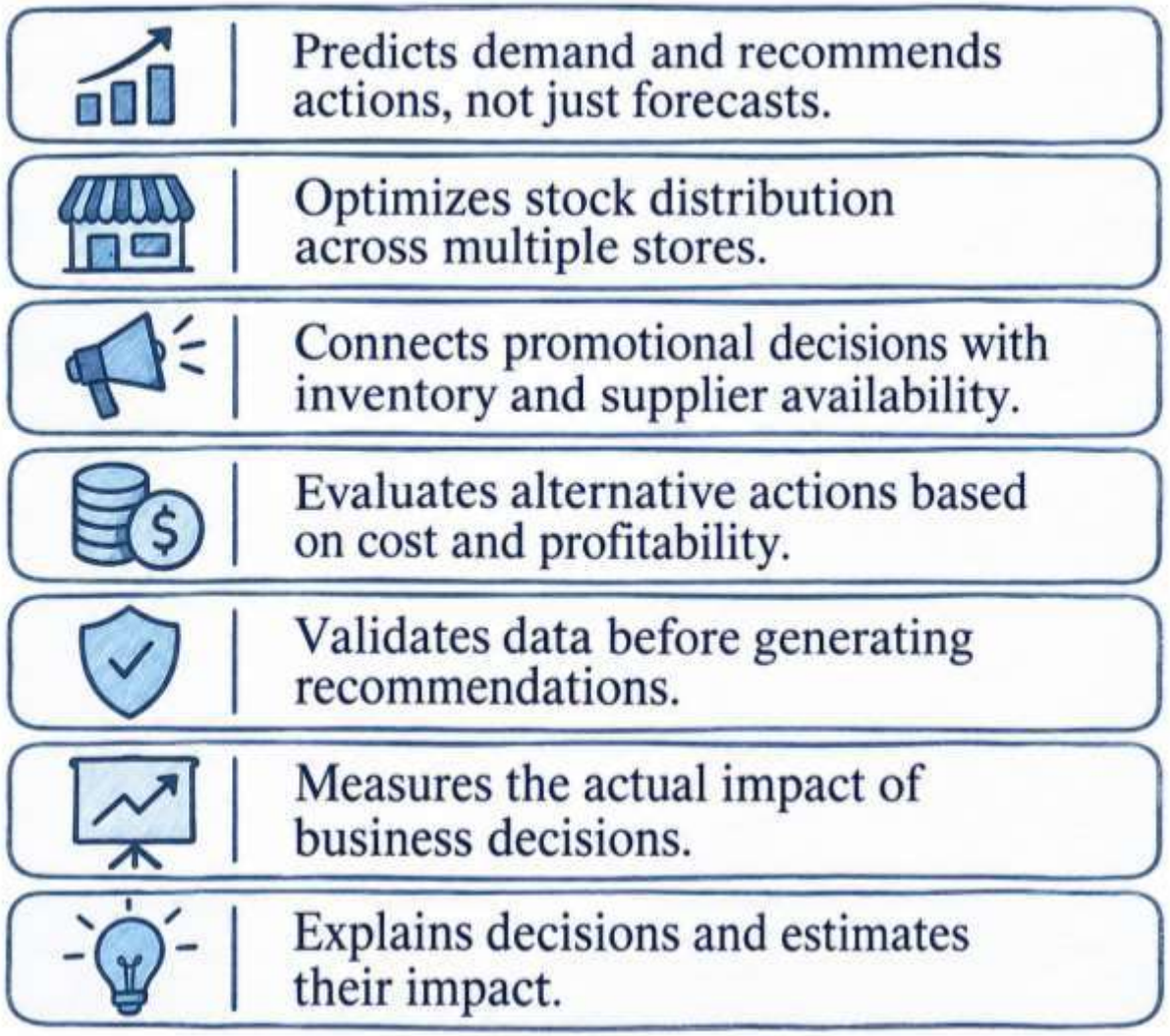
KINTRAL – CENTRALZED PLATFORM

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Proposed Solution

- **Centralized Retail Data Monitoring:** Collects and validates sales, inventory and supplier data from multiple stores.
- **Demand Forecasting & Risk Prediction:** Predicts demand and identifies stock shortages, excess stock and expiry risks.
- **Intelligent Decision & Optimization Engine:** Evaluates stock transfers, replenishment, supplier alternatives and discount options.
- **Explainable Recommendations & Impact Analysis:** Suggests the best action with reasons, costs and expected benefits.
- **Action Execution & Performance Measurement:** Allows managers to approve or simulate actions and measure their business impact.

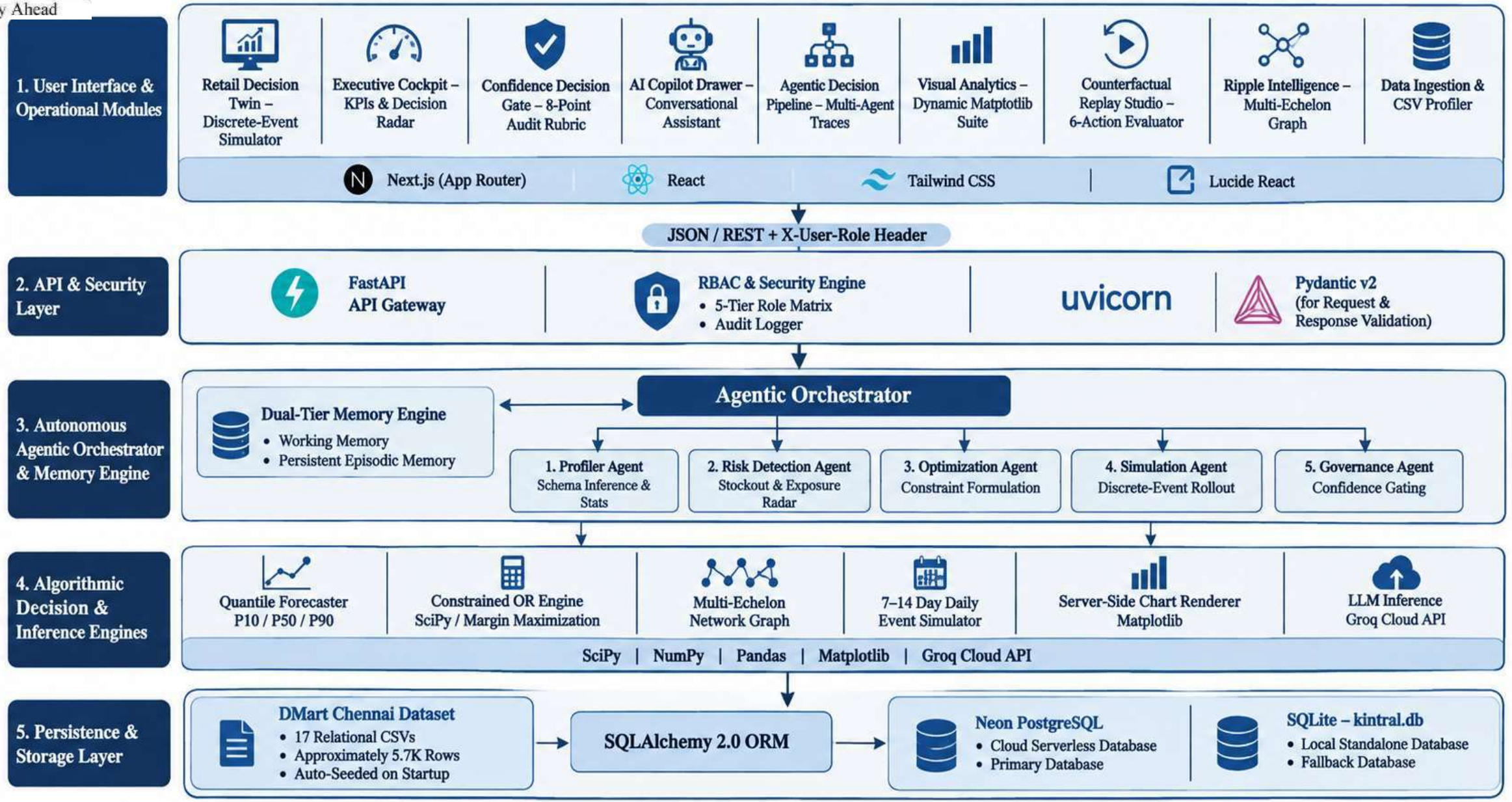
Innovation and Uniqueness



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TOOLS AND TECHNOLOGIES

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IMPACT AND FEASIBILITIES

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Technical Feasibility

- Existing AI/ML technologies and open-source tools.
- Integration of demand forecasting and decision engines.
- Interactive dashboard for retail analytics and recommendations.

Data Feasibility

- Availability of historical sales and inventory datasets.
- Supplier, pricing, promotion and stock-transfer information.
- Continuous collection and validation of retail data.

Operational Feasibility

- Simple daily sales and inventory data entry.
- Easy monitoring of multiple stores and products.
- Manager approval and simulation of recommended actions.

Economical Feasibility

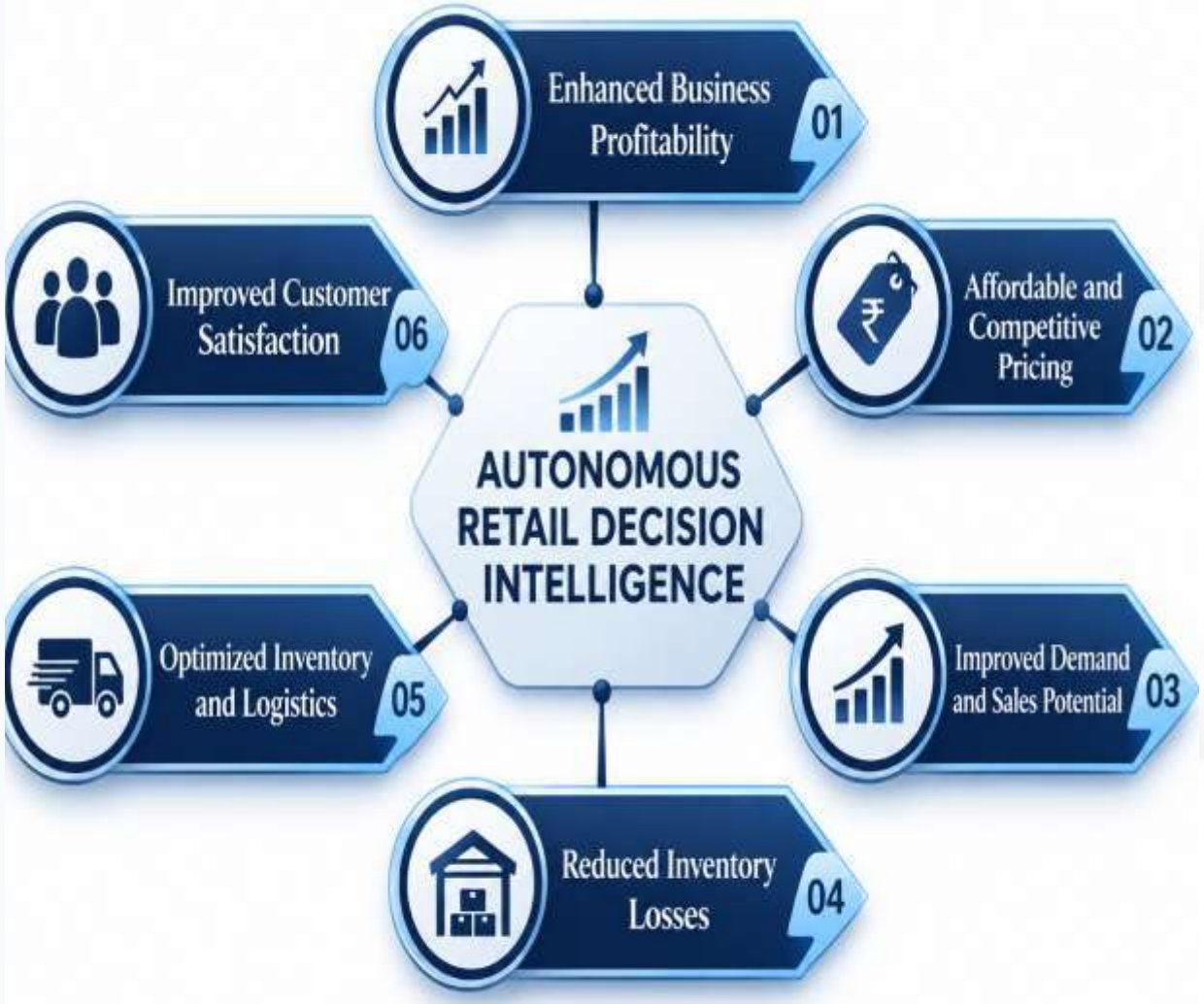
- Low development cost using open-source technologies.
- No specialized hardware dependency.
- Potential reduction in inventory losses and operational costs.

Scalability Feasibility

- Supports multiple stores and thousands of SKUs.
- Expandable to additional retail branches and distribution centres.
- Supports integration of advanced forecasting and optimization models.

User Feasibility

- User-friendly dashboard for different retail managers.
- Visual reports, alerts and actionable recommendations.
- Easy-to-understand explanations for AI-generated decisions.



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FUTURE SCOPE AND SCALABILITY

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